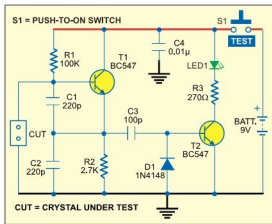


Simple Crystal Tester

VENUGOPAL S.

Here is a simple circuit that can be used to test a crystal before using it in a circuit. The circuit is built around two BC547 transistors (T1 and T2) and a few discrete components.

The oscillator circuit formed by transistor T1, resistors R1 and R2, and capacitors C1 and C2 oscillates if a good crystal is connected to the test points marked as CUT (crystal under test). The output from the oscillator is rectified by diode D1 and filtered by capacitor C3. The positive voltage appearing across the capacitor is fed to the base of transistor T2, causing it to conduct.



Testing of a crystal is simple: Insert the crystal at CUT points shown in the circuit diagram and press test switch S1. If LED1 glows, your crystal is good

and you can use it in a circuit.

The circuit is powered by a standard 9V battery. Push-to-on switch S1 is included to prolong the battery life but it's not needed if you use a socket for the crystal under test.

Assemble the circuit on a general-purpose PCB and enclose in a suitable small cabinet. Fix the two-pin connector, LED1 and test switch on top of the cabinet. Fix the 9V battery inside the cabinet.

Note. You can use 2N3563 transistors instead of BC547. ●



Touch-Free Timer Switch

T.K. HAREENDRAN

This type of infrared proximity circuit is widely used as an electric switch where physical contact is not desired for hygiene purpose. For example, we commonly see use of infrared proximity sensors on public drinking fountains and in public wash-

rooms. The simple circuit presented here can be operated by moving your hand in front of it. This is achieved by detecting the infrared light reflected by your hand onto a receiver diode.

Fig. 1 shows the circuit of the touch-free timer switch. It has two sections: transmitter and receiver. The IR transmitter is built around timer

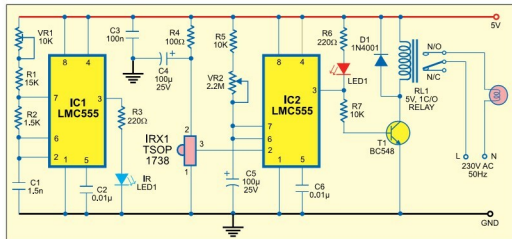


Fig. 1: Circuit of touch-free timer switch

LMC555 (IC1), which is wired as an astable multivibrator. The multivibrator produces 38kHz pulses (at low duty cycle) that drive an infrared LED (LED1). This frequency can be tuned using a 10-kilo-ohm preset (VR1). A 220-ohm series resistor (R3) ensures that the current consumption of the IR transmitter is not out of arrangement.

The receiver section is built around IR receiver module TSOP1738 (IRX1), timer LMC555 (IC2) and a few discrete components. The TSOP1738 is an integrated miniaturised receiver for infrared remote control systems. Everything required for IR signal processing, including the PIN diode and preamplifier, are assembled on a lead frame

